ModelIt!

Teacher's Guide

What Makes a Shadow?

The Science of Light and Shadows

Quick Reference

Grade Level	Grade 1
NGSS Standard	1-PS4-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 1-PS4-3

Plan and conduct investigations to determine the effect of placing objects made with different materials in the path of a beam of light.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students investigate how changing the position of a light source affects the size and shape of shadows.
Disciplinary Core Idea	**PS4.B Electromagnetic Radiation** — Objects can be seen only when light is available to illuminate them; some materials block light and create shadows.
Crosscutting Concept	**Cause and Effect** — Students explore how changing the light position causes the shadow size to change.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Shadows are everywhere! Every time light hits an object that blocks it, a shadow is created. Understanding shadows helps us learn how light works and how it travels.

How Shadows Form

Light travels in straight lines. When light hits a transparent material like glass, it passes right through. But when light hits an opaque material like cardboard or your body, it cannot get through. The area behind the object is dark because the light is blocked — this dark area is a shadow. Every shadow needs three things: a light source, an object to block the light, and a surface for the shadow to fall on. Without any one of these three, you cannot make a shadow.

Why Shadows Change Size

The size of a shadow depends on where the light source is. When the light is CLOSE to the object, the shadow is very BIG because the object blocks a large portion of the light. When the light is FAR from the object, the shadow is SMALLER because the object blocks a smaller portion of the total light. You can see this in real life — your shadow is very long early in the morning when the sun is low and close to the horizon, and very short at noon when the sun is high overhead. Shadow puppeteers use this trick to make their puppets look huge on the wall!

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify light position as the external component they control (moving the flashlight closer or farther) and shadow size as the internal result that changes.

E **Establish Relationships**
Students discover that when the light gets closer, the shadow gets bigger — and when the light moves away, the shadow gets smaller.

V **Visualize & Model**
Students build a simple model showing how light position connects to shadow size.

E **Evaluate Outcomes**
Students test their model by moving a flashlight closer and farther from an object and measuring the shadow.

R **Revise & Extend**
Students reflect on how their model explains why shadows are long in the morning and short at noon.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Have you ever made a shadow puppet on the wall? How did you do it? What did you need?"*

Watch For

Turn off some lights and use a flashlight to make a shadow on the wall. Ask: What three things do we need to make a shadow?

Slide 2: What Will We Learn?

3 min

Say: *"Today we will learn what makes shadows and why they change size! We will learn two words: SHADOW and LIGHT SOURCE."*

Watch For

Ask: What are some light sources you know? (sun, flashlight, lamp) Practice saying shadow and light source together.

Slide 3: Let's Be Scientists!

3 min

Say: *"What makes a shadow appear, and why do shadows change shape? Have you noticed your shadow looks different at different times?"*

Watch For

If possible, go outside briefly and observe shadows. Or show photos of shadows at morning vs. noon.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts: Light Position is where we put the flashlight. Shadow Size is how big the shadow gets."*

Watch For

Place a toy on a table. Shine a flashlight from different positions. Ask: What happens to the shadow?

Slide 5: How Are They Connected?

5 min

Say: *"When the light moves closer, does the shadow get bigger or smaller? Closer means BIGGER! Can anyone tell me why?"*

Watch For

Let students move the flashlight closer and farther to see the effect. Use sentence frame for the connection.

Slide 6: What Did We Find Out? 5 min

Say: *"We learned that shadows happen when something blocks light. The closer the light, the bigger the shadow. The farther the light, the smaller the shadow."*

Watch For

Review all discoveries. Challenge students: Can you make a shadow with a clear cup? (No! Light goes through it.)

Slide 7: STEM Challenge 10 min

Say: *"Time for a shadow puppet show! Cut out shapes, attach them to sticks, and put on a show. Make your puppets look big and small!"*

Watch For

Distribute cardboard, scissors, popsicle sticks, tape, and flashlights. Students create shadow puppets and perform short shows.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Light Position
- **INTERNAL:** Shadow Size

Relationships

- **Light Position to Shadow Size:** POSITIVE (+) — When the light moves closer to the object, the shadow gets bigger because the object blocks a larger area of the light. Moving the light farther away makes the shadow smaller.

Key Discoveries

- Shadows appear when an object blocks light from passing through
- When the light is closer to the object, the shadow gets bigger
- When the light is farther from the object, the shadow gets smaller
- Shadows always appear on the opposite side of the object from the light

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Shadows are alive or are part of you"	Shadows are not alive and they are not a piece of you. A shadow is just a dark area where light cannot reach because your body is blocking it. If you move, the shadow moves because the blocked area moves. If the light goes away, the shadow disappears completely.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Dark objects make darker shadows than light objects"	The color of an object does not change how dark a shadow is. A white ball and a black ball both make the same shadow because shadows are created by BLOCKING light, not by color. What matters is whether the object is opaque — whether it blocks light. A clear object makes no shadow at all, regardless of its color.